20 Years of the Double Pukelsheim System in Switzerland: What Is the Mechanical Effect of the Transition from the Hagenbach-Bischoff Method on the Seat Distribution of Smaller Parties?

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Abstract

Here I write normal text.

Introduction

Theory/Literature Review

Hypotheses

Analysis

Data Cleaning

To be able to analyze the data from the different elections and compare the outcomes. each dataset has to be cleaned. The data For some cantons e.g. Valais and Schwyz the data is very scattered, therefore for there multiple datasets per election have to be cleaned and merged.

The procedure for the data cleaning was the following: For each canton in alphabetically order, for each election the data was cleaned.

To ensure the compatibility of the different datasets, the column names were standardized. The column names were renamed for each dataset to the following: “total_votes”, “svp_votes”, “sp_votes”, “cvp_votes”, “fdp_votes”, “glp_votes”, “bdp_votes”, “green_votes”, “other_votes”. It might be that some of the datasets do not contain all of these columns, since the party landscapes varies geographically and over time.

For the counterfactual, the district-level party votes and district seat magnitudes from the pre-Pukelsheim era are retained. The Hagenbach-Bischoff allocation follows the relevant cantonal law of the time (e.g., Sections 12-13 of the Aargau Grossratswahlgesetz for the 2005 election). The main counterfactual treats every party independently, ignoring list alliances, as they were abolished with the transition to Double Pukelsheim. Vote shares use each canton’s *Wählerzahl*

definition: district-level party votes are divided by the district seat magnitude and rounded to the nearest integer before cantonal aggregation.

The official Double Pukelsheim results (`seats_dp`) are sourced from official cantonal authorities and are documented in the Bibliography. A single lookup table consolidates all official results for easy merging with the Hagenbach-Bischoff counterfactuals. All data processing is implemented in canton-specific R scripts (R/canton_*.R) which apply shared functions for algorithm execution and validation.

Results

Interpretation of the Results

Discussion

Conclusion

Bibliography

Department für Finanzen und Ressourcen des Kantons Aargau. (2009, April 21). *Detailergebnisse der Grossratswahlen vom 8. März 2009*. https://www.ag.ch/de/medien/medienmitteilungen?mm=detailergebnisse-der-grossratswahlen-vom-8-maerz-2009-742f7d21-d4b0-454f-abd7-25471ef7f566_de

Kanton Aargau. (2004). *Grossratswahlgesetz: Änderung vom 8. Juni 2004*. Abstimmungsbroschüre zur kantonalen Volksabstimmung vom 26. September 2004, pp. 5–8. https://www.ag.ch/media/kanton-aargau/portal/aktuell/wahlen-abstimmungen/abstimmungen/abstimmungsbroschueren/20040926-abstimmungsbroschuere.pdf

Kanton Aargau. (2008). *Gesetz über die Wahl des Grossen Rates (Grossratswahlgesetz), Fassung gemäss Gesetz vom 18. September 2007, in Kraft seit 1. Juli 2008*. https://gesetzessammlungen.ag.ch/api/de/versions/1234/pdf_file

Staatskanzlei des Kantons Aargau. (2008, September 15). *Grossratswahlen: Vorverfahren gestartet*. https://www.ag.ch/de/medien/medienmitteilungen?mm=Grossratswahlen-Vorverfahren-gestartet-50a84551-3d6d-454c-a33c-5644c8bf50a5_de

Staatskanzlei des Kantons Aargau. (2005, 7. März). Bezirksprotokolle über die Erneuerungswahl des Grossen Rats für die Amtsperiode 2005/2009: Wahltag: 27. Februar 2005 (Amtsblattbeilage 2005-1). https://doczz.net/doc/5954970/amtsblattbeilage_vom_07.03.2005

Appendix

Appendix 1: Explanation Data Cleaning Process

Appendix 2: Code for Data Cleaning

Declaration of AI-Assistance